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Edge cloud resource allocation using hints from network providers

Abstract

Aspects of the subject disclosure may include, for example, a network API service that receives network event data and provides performance hints to a resource manager that manages application containers at edge cloud locations. Network event data may be received from access networks, core networks, nodes within access networks or core networks, or the like. Performance hints may allow booting of application containers at edge cloud locations. Other embodiments are disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims priority to U.S. application Ser. No. 18/298,026, filed Apr. 10, 2023, which is a continuation of U.S. application Ser. No. 17/751,730, filed May 24, 2022, now U.S. Pat. No. 11,652,756, which is a continuation of U.S. application Ser. No. 17/231,721, filed Apr. 15, 2021, now U.S. Pat. No. 11,374,875. All sections of the aforementioned applications and patents are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) The subject disclosure relates to allocating edge cloud resources.

BACKGROUND

(2) An edge cloud platform may include of a number of small data centers deployed at hundreds of

locations within a network. Resource allocation is challenging at edge cloud locations because of their small size and the variety of applications to be supported.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
- (2) FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.
- (3) FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a network application programming interface (API) service functioning within the communication network of FIG. 1 in accordance with various aspects described herein.
- (4) FIG. 2B is a block diagram illustrating an example, non-limiting embodiment of an authentication method in accordance with various aspects described herein.
- (5) FIG. 2C is a block diagram illustrating an example, non-limiting embodiment of a network API service providing guidance functionality in accordance with various aspects described herein.
- (6) FIG. 2D is a block diagram illustrating an example, non-limiting embodiment of a distributed network API service in accordance with various aspects described herein.
- (7) FIGS. 2E-2G depict illustrative embodiments of methods in accordance with various aspects described herein.
- (8) FIG. 2H is a block diagram illustrating an example, non-limiting embodiment of a resource manager interacting with a network API service in accordance with various aspects described herein.
- (9) FIG. 2I is a block diagram illustrating an example, non-limiting embodiment of an application container at a first edge cloud location in accordance with various aspects described herein.
- (10) FIG. 2J is a block diagram illustrating an example, non-limiting embodiment of an application container moving from a first edge cloud location to a second edge cloud location in accordance with various aspects described herein.
- (11) FIG. 2K is a block diagram illustrating an example, non-limiting embodiment of an edge cloud platform locating an application server in an edge cloud location that communicates through a local breakout function in accordance with various aspects described herein.
- (12) FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.
- (13) FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.
- (14) FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.
- (15) FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

- (16) The subject disclosure describes, among other things, illustrative embodiments for managing edge cloud resources in response to network events. Other embodiments are described in the subject disclosure.
- (17) One or more aspects of the subject disclosure include a device having a processing system including a processor and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations. The operations may include receiving a request for edge cloud related network hints from an edge cloud resource manager, wherein the edge cloud related network hints are specific to a user equipment (UE) device, receiving network

event data related to the UE device, creating the edge cloud related network hints from the network event data, and providing the edge cloud related network hints to the edge cloud resource manager. (18) Additional aspects include providing an authentication nonce to the UE device, wherein the request for the edge cloud related network hints includes the authentication nonce; providing the edge cloud related network hints to an edge application master node; providing the edge cloud related network hints to an edge cloud platform; receiving network event data generated when the UE device attaches to, or detaches from, a network; and providing hints to boot or migrate an application container.

(19) One or more aspects of the subject disclosure include a non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations. The operations may include receiving a request for edge cloud related network hints from an edge cloud resource manager, wherein the edge cloud related network hints are specific to a user equipment (UE) device, receiving network event data related to the UE device, creating the edge cloud related network hints from the network event data, and providing the edge cloud related network hints to the edge cloud resource manager.

(20) One or more aspects of the subject disclosure include a method comprising receiving, by a processing system including a processor, a request for edge cloud related network hints from an edge cloud resource manager, wherein the edge cloud related network hints are specific to a user equipment (UE) device receiving, by the processing system, network event data related to the UE device, creating, by the processing system, the edge cloud related network hints from the network event data, and providing, by the processing system, the edge cloud related network hints to the edge cloud resource manager.

(21) Additional aspects include devices, non-transitory machine readable medium, and methods for providing an authentication nonce to the UE device, wherein the request for the edge cloud related network hints includes the authentication nonce; providing the edge cloud related network hints to an edge application master node; providing the edge cloud related network hints to an edge cloud platform; receiving network event data generated when the UE device attaches to, or detaches from, a network; and providing hints to boot or migrate an application container.

(22) Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, system **100** can facilitate in whole or in part a network API service that combines data from access and core networks and provides guidance and control services to API users. In particular, a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

(23) The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VOIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

(24) The communications network **125** also includes network API service **158**. As further described below with reference to later figures, network API service **158** receives network event data from various network elements and access networks, and then provides hints to various entities to enable application container provisioning in the cloud. For example, network API service **158** may register to receive network event data from access networks such as wireless access **120**, broadband access **110**, voice access **130**, media access **140**, and the like. Also for example, network API service **158** may register to receive network event data from network elements within a core network such as NEs **150**, **152**, **154**, and **156** within communications network **125**.

(25) In some embodiments, network API service **158** may register to receive network event data from various sources. For example, network API service **158** may receive network event data from multiple access networks (e.g., cellular, fixed-access) as well as core networks (e.g., mobility core, IP backbone). Examples of network event data include data reflecting messages that pass between the UE and a radio access network, between a UE and a core network, between a radio access network and a core network, and the like. In some embodiments, the network event data provides information alerting network API service **158** that a UE is attaching to a radio access network, detaching from a radio access network, or being handed over from one node to another node in a radio access network. Any available network event data may be utilized as described herein. Network API service **158** may also provide a predictor that ingests data streams of network event data from multiple sources and outputs edge cloud related hints to allow an edge cloud resource manager to boot application containers in an efficient manner. In some embodiments, the edge cloud resource manager may be an edge application master node that manages containers and application instances for a particular software application to serve the UE, and in other embodiments, the edge cloud resource manager may be a resource manager controlled by the edge cloud provider. These and other embodiments are more fully described below with reference to later figures.

(26) In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

(27) In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

(28) In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

(29) In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway, or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

(30) In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

(31) In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs,

servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

(32) FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a network application programming interface (API) service functioning within the communication network of FIG. 1 in accordance with various aspects described herein. FIG. 2A shows network API service **158** communicating with user equipment (UE) **210A**, access network **220A**, core network **230A**, and server application **240A**. UE **210A** may be any device capable of communicating with access network **220A** and network API service **158**. For example, UE **210A** may be cellular user equipment, set top boxes or routers, or other customer premise equipment. Also for example, UE **210A** may be an Internet of Things (IoT) device, such as a smart appliance, a connected water or power meter, a camera, or the like. Also for example, UE **210A** may be limited to a single application or program being executed on a user device. An example application may be an app on a smartphone, a video application on a smart TV, or a navigation app in an automobile. In some embodiments, UE **210A** may support multiple access networks. For example, UE **210A** may be capable of connecting to a cellular radio access network, a Wi-Fi access point, a tethered broadband connection, or any other type of access network. The terms “user device” and “user equipment” are used interchangeably herein.

(33) Access network **220A** may be any type (or any number) of access networks that enables UE **210A** to communicate with core network **230A**. For example, access network **220A** may be any of the access networks described above with reference to FIG. 1. Access network **220A** may include many devices or elements capable of providing data to network API service **158** at **222A**. For example, access network **220A** may include network routers, switches, cellular base stations, and the like.

(34) Core network **230A** may be any communications network capable of providing services to UE **210A** through an access network, such as access network **220A**. For example, core network **230A** may be implemented as communications network **125** (FIG. 1). In some embodiments, core network **230A** provides an entry point to data networks for user devices. For example, core network **230A** may access data networks outside of core network **230A** to provide connectivity between external networks and UE **210A**. Also for example, core network **230A** may provide connectivity to a server such as server **240A**.

(35) In some embodiments, server **240A** may be a hardware server upon which applications may execute, where those applications provide services to UE **210A**. Also in some embodiments, server **240A** may be a virtual machine or virtual network element executing on hardware within core **230A**, in a cloud environment, or in a distributed environment.

(36) In some embodiments, network API service **158** may receive network event data from one or more of the elements shown in FIG. 2A, predict UE behavior based on the network event data, and may then provide performance hints to one or more of the same elements shown FIG. 2A. For example, network API service **158** may receive network event data from access network **220A** at **222A** and/or core network **230A** at **232A**. Examples include current attachment events and messages, detachment events and messages, messages signaling UE handover events, and the like.

(37) In some embodiments, network elements within core network **230A** and elements within access network **220A** offer services to provide network event data, and network API service **158** may subscribe to these services. In some embodiments, elements within access network **220A** and core network **230A** may output streams of data or events on a per-user basis or aggregate basis with on/off subscriptions. Data may be collected from these elements on a periodic basis or an on-demand basis. For example, in some embodiments, network event data is collected in real-time as it is generated.

(38) In some embodiments, network API service **158** responds to requests from server **240A** to

provide performance hints based on the network event data. For example, in some embodiments, network API service **158** may analyze and/or synthesize the network event data, predict operations that may enhance performance of edge cloud entities, and then provide performance hints server **240A**. Performance hints may include identification of edge cloud locations at which an application container should be instantiated, commands to boot an application container or virtual machine at a particular edge cloud location, commands to take down an application container or virtual machine at an edge cloud location, or the like.

(39) FIG. **2B** is a block diagram illustrating an example, non-limiting embodiment of an authentication method in accordance with various aspects described herein. In some embodiments, network API service **158** requires users of the API service to first be authenticated. The example authentication process shown in FIG. **2B** authenticates UE **210A** to network API service **158** and then verifies that future API service requests are made by authenticated users or by other elements on behalf of the authenticated users.

(40) UE **210A** provides an authentication request to network API service **158** at **210B**. In response to the authentication request, network API service **158** determines whether UE **210A** is provisioned for network API service privileges. In some embodiments, this may be limited to verifying that UE **210A** is a mobility customer of a network operator. In other embodiments, this may include verifying that a user profile includes information specifying that the user has network API service privileges.

(41) The authentication request made at **210B** may be performed on behalf of a user, a specific user device, or a specific user application executing on a specific device. For example, when the authentication is on behalf of a user, then that authenticated user may request network API services from any device or any application. Also for example, when the authentication is on behalf of a user device, then that user device may request network API services on behalf of any application running on the user device. Further, when the authentication is on behalf of an application, then API services may be limited to requests made solely by that application. In general, authentication may be performed at any level of granularity and on behalf of any person, hardware device, or process executing on a hardware device.

(42) In response to determining that UE **210A** has network API service privileges, network API service **158** provides an authentication nonce to UE **210A** at **212B**. In some embodiments, the authentication nonce is a one-time-use identifier that uniquely identifies UE **210A** as having been authenticated by network API service **158**. The authentication nonce may be randomly generated, cryptographically generated, or may be generated using any authentication process. In some embodiments, the authentication nonce represents not only an identifier, but a process. For example, the authentication nonce may be a shared key, a public key, or a cryptographic challenge/response.

(43) Once UE **210A** is authenticated, UE **210A** may make an API request at **220B**. As part of the API request, UE **210A** provides the authentication nonce to network API service **158**. Network API service **158** verifies the authentication nonce as valid and provides an API response to UE **210A** at **222B**. The interactions between UE **210A** and network API service **158** shown at **210B**, **212B**, **220B**, and **222B** represent a user authenticating to a network API service and the network API service limiting access to a user that has been previously authenticated. In some embodiments, network API service **158** will respond to requests from elements that have not been directly authenticated, but have been authorized by authenticated users. These and other embodiments are described further below.

(44) In some embodiments, an authenticated user may provide an authentication nonce to a separate element to authorize that separate element to access network API services on behalf of the authenticated user. For example, as shown in FIG. **2B**, UE **210A** may provide the authentication nonce to server **240A** at **230B**. Server **240A** may perform an API request at **240B**. The API request at **240B** may include the authentication nonce provided by UE **210A**. In response to the API

request at **240B**, network API service **158** verifies the authentication nonce as valid and provides an API response to server **240A** at **242B**.

(45) In some embodiments, authentication to network API service **158** is limited to users such as UE **210A**, and does not allow direct authentication by other elements such as server **240A**. In other embodiments, elements other than users may be provided direct authentication methods thereby allowing a server such as server **240A** to provide its own authentication nonce when performing an API request.

(46) FIG. 2C is a block diagram illustrating an example, non-limiting embodiment of a network API service providing guidance functionality in accordance with various aspects described herein. As shown in FIG. 2C, network API service **158** includes predictor **220C** and network API gateway **230C**. In some embodiments, predictor **220C** receives network event data from access network **220A** and/or core network **230A**, and predicts actions that will enhance performance based on the received network event data. For example, predictor **220C** may receive network event data from access network **220A** alerting predictor **220C** that UE **210** is beginning an attach procedure. In response to this network event data, predictor **220C** may predict that booting an application container at a specific edge cloud location may enhance performance.

(47) Predictor **220C** may be implemented in any manner. For example, predictor **220C** may be a process or program running on a virtual machine or server, and may also be implemented in a distributed manner. Predictor **220C** may include storage or may be stateless. For example, in some embodiments, predictor **220C** may analyze received network event data without providing for long term storage for the results.

(48) Network API gateway **230C** provides an external interface for API network service **158**. For example, network API gateway **230C** may provide a list of APIs available to applications, provide a mechanism to register applications and authenticate users, and to request user consent to release data.

(49) In operation, network API service **158** receives network event data from various elements as shown in FIG. 2C. For example, access network **220A** may provide network event data at **222C** and core network **230A** may provide network event data at **232C**. The network event data may be collected periodically or in response to events. For example, access network **220A** may provide network event data when a particular user authenticates to the access network. Also for example, network API service **158** may subscribe to one or more services published by elements within access network **220A** to receive network event data either on a periodic basis or as a result of network events. Similarly, core network **230A** may provide network event data either on a periodic basis or as a result of network events. Various network elements within core network **230A** may publish services to provide network event data, and network API service **158** may subscribe to those services within core network **230A** to receive network event data either on a periodic basis or as a result of network events.

(50) Predictor **220C** may provide any type of analytics or synthesis in support of the operation of network API service **158**. For example, predictor **220C** may consume raw data streams from network elements within access network **220A** and/or core network **230A** to produce performance hints in real time. Also for example, in some embodiments, predictor **220C** may perform future predictions from past data. Examples include predictions of particular applications to be used by UE **210A**, particular locations which UE **210A** may occupy, and likely future handoff scenarios. In some embodiments, predictor **220C** may provide user specific predictions or provide predictions for classes of users (e.g., users in a given location, time of day, device type, etc.). Also for example, predictor **220C** may keep historical records of predictions to be used for future predictions. The historical records may be maintained on an aggregate basis, on a per-user basis, or any other basis.

(51) Network API gateway **230C** may provide authentication services to applications, user devices, or users. For example, as discussed above with reference to FIG. 2B, network API gateway **230C** may receive authentication requests, determine that the requestor has sufficient privileges, and then

provide an authentication nonce to the requestor. Network API gateway **230C** may also receive requests for API services.

(52) In some embodiments, network API gateway **230C** may also receive API requests from entities that have not been directly authenticated. For example, network API gateway **230C** may receive an API request from server **240A**. In some embodiments, server **240A** may provide, as part of the API request, an authentication nonce previously registered to a user device. In other embodiments, server **240A** may identify UE **210A** in some other manner as part of the API request. In these embodiments, network API gateway **230C** may request permission from UE **210A** to respond to server **240A**. UE **210A** may grant or deny the permission. When server **240A** has the appropriate permission, via an authentication nonce or other mechanism, network API gateway **230C** utilizes predictor **220C** to respond to the API request.

(53) In some embodiments, server **240A**, after receiving performance hints from network API service **158**, may perform server-side adaptations. For example, server **240A** may boot an application container in a specific edge cloud location in response to performance hints received from network API service **158**. In general, server **240A** may take any action or perform any adaptation in response to performance hints received from network API service **158** in response to API requests.

(54) FIG. 2D is a block diagram illustrating an example, non-limiting embodiment of a distributed network API service in accordance with various aspects described herein. The network API service shown in FIG. 2D is distributed across three geographical regions. For example, network API service **212D** is located in geographical region one, network API service **222D** is located in geographical region two and network API service **232D** is located in geographical region three. In some embodiments, the different network API services are implemented in a network core or on network edges, or in cloud-based network edge locations. In some embodiments, users in a particular geographical region are redirected to network API services that are physically located in the same geographical regions when requesting network API services. For example, user **210D** may make a request of a centralized network API service such as network API service **158** (FIG. 2A), and user **210D** may be redirected to network API service **212D** to reduce latency and network bandwidth consumption. Similarly, user **220D** may be redirected to network API service **222D** and user **230D** may be redirected to network API service **232D**.

(55) In some embodiments, network API services in different geographical regions subscribe to the same services to receive identical network event data. In these embodiments, any network API service in any geographical region may have access to the same network event data regardless of user location or API service location. In other embodiments, distributed network API services emphasize the collection of network event data describing network events in their geographical region. In these embodiments, network API services in different geographical regions may have partial overlap of network event data collected and analyzed, but may also receive network event data unique to their geographical region. In these embodiments, a network API service in a particular geographical region may be more likely to receive network event data of interest to a user in the same geographical region.

(56) In some embodiments, a single server such as server **240A** in FIG. 2D may communicate with multiple users in different geographic regions. In these embodiments, this single server may communicate with multiple network API services in different locations depending on the geographical regions in which the users operate. For example, if user **210D** has authenticated and received an authentication nonce from network API service **212D** and provided that authentication nonce to server **240A**, server **240A** may provide that authentication nonce to network API service **212D** when requesting performance hints on behalf of user **210D**. Likewise, server **240A** may provide an authentication nonce received from user **220D** when communicating with network API service **222D**, and may use an authentication nonce received from user **230D** when communicating with network API service **232D**.

(57) FIGS. 2E-2G depict illustrative embodiments of methods in accordance with various aspects described herein. Referring to FIG. 2E, method **200E** may be performed by a user, a user equipment (UE) device, an application running on a UE device, or any other consumer of network API services. For example, the operations of method **200E** may be performed by a broadband customer having a tethered connection to a laptop computer, a mobility customer with a handheld device connected to a radio access network, an application running on an infotainment center in an automobile, or the like.

(58) At **210E**, a request is made to authenticate a UE device to a network API service. In some embodiments, this corresponds to a UE device requesting authentication from a network API service such as network API service **158**. At **220E**, an authentication nonce is received. In some embodiments, the authentication nonce is a cryptographic nonce, and in other embodiments the authentication nonce is a one-time-use randomly generated number used as a user-ID. In some embodiments, the authentication nonce is provided only after the UE device has been verified as a device with network API service privileges.

(59) At **230E**, the authentication nonce is provided to a server to enable the server to request performance hints from the network API service. In some embodiments, this allows an edge application resource management server to request guidance for management edge cloud resources on behalf of the UE device that was previously authenticated to the network API service. In some embodiments, the UE device performing method **200E** may receive, from a network API service, a request to allow a server to access the network API service on behalf of the UE device. In these embodiments, the UE device may grant or deny the permission based on any criteria.

(60) Referring to FIG. 2F, method **200F** may be performed by a network API service such as network API service **158**. At **210F**, network event data is received. In some embodiments, the network event data is collected from access networks, core networks, or any other network element in communication with the network API service.

(61) Example network event data received from an access network may include Radio Resource Control (RRC) connection establishment/RRC idle (another link). For example, messages that may give rise to network data events being reported to a network API service include RRC Connection Request (SRB0, CCCH/UL_SCH/PUSCH), RRC connection setup (SRB0, CCCH/DL-SCH/PDSCH) and RRC Connection SETUP Complete (SRB1, DCCH/UL-SCH/PUSCH). These and other events may include a UE identity to uniquely identify a user, a radio resource identification, or other information useful to determine a UE, a UE location, a UE state of attachment, a radio resource, a radio resource location or the like. In 4G embodiments, when a UE goes from RRC_IDLE to RRC_connected or vice versa, the eNB may report the event to the network API service. Further, in 5G embodiments, transitions between RRC_IDLE, RRC_INACTIVE, and RRC_CONNECTED may be reported to the network API service. In some embodiments, access network events may be received earlier than core network events which may give more time to spin up containers.

(62) Example core network events include attach and detach events and/or handover events. In some embodiments, attach and detach events may include an initial UE attach, and initial context setup request, and/or a UE context release. Also for example, a handover event may include a tracking area update (e.g., a change in UE location), a mobility management unit (MME) triggered serving gateway relocation (e.g., switching from serving or packet gateway to another serving or packet gateway). Network event data may be received from any type of network core. For example, in some embodiments, events may be received from a 4G MME, and in other embodiments, events may be received from a 5G access and mobility management function (AMF).

(63) In some embodiments, messages passed between a UE, an access network and/or a core network are reported directly to a network API service as network data events. In other embodiments, the network data events reported to the network API service include information related to, or derived from, the messages passed between a UE, an access network and/or a core

network. For example, in some embodiments, a radio access network may pass the contents of an RRC connection request to a network API service as a network data event. Also for example, in some embodiments, a radio access network may pass only a portion an RRC connection request to a network API service as a network data event. Also for example, in still further embodiments, a radio access network may pass only a UE identifier, a radio resource identifier, and a connection state to a network API service as a network data event in response to receiving an RRC connection request.

(64) At **215F**, a request to authenticate a UE device is received at the network API service. In some embodiments, this corresponds to a request being received on behalf of a user, on behalf of a UE device, or on behalf of an application running on a UE device. For example, the authentication request may provide authentication to all devices registered to a particular user, to the particular device that is communicating the request for authentication, or to a particular application that is running on the UE device requesting authentication.

(65) At **220F**, the network API service determines that the UE device has network API service privileges. In some embodiments, this corresponds to verifying that the UE device is registered to a user of a particular service provider. In other embodiments, this corresponds to verifying that a user account corresponding to the UE device or the application requesting authentication has specific privileges to access one or more network API services. At **225F**, in response to determining that the UE device has network API service privileges, an authentication nonce is provided to the UE device. In some embodiments, the authentication nonce is a cryptographic nonce, and in other embodiments, the authentication nonce is non-cryptographic.

(66) At **250F**, the authentication nonce previously provided to the UE device at **225F** is received from a server with a request for performance hints. In some embodiments, this corresponds to a server having received a copy of an authentication nonce from an authenticated user so that the server may request performance hints from a network API service on behalf of the user. At **255F**, network event data is analyzed and/or synthesized to predict performance hints. At **260F**, the performance hints predicted at **255F** is provided to the requesting server. The requesting server may perform adaptations based on the performance hints provided at **260F**. For example, the server application may select an edge cloud location to house an application container, boot an application container, or take down an application container in response to performance hints provided at **260F**.

(67) Referring to FIG. 2G, method **200G** may be performed by a server such as an edge cloud management server. For example, a server running as a virtual machine, or an application running on a virtual machine may perform the various actions of method **200G**. At **210G**, an authentication nonce is received from a UE device. In some embodiments, this corresponds to a UE device providing an authentication nonce to the server so that the server may request network API services on behalf of the user device. At **220G**, the authentication nonce is provided to a network API service to request performance hints. At **230G**, responsive to performance hints received at the server, the server performs an adaptation. For example, the server may select an edge cloud location to house an application container, boot an application container, or take down an application container in response to performance hints provided at **230G**. In response to received performance hints, a server may create or destroy containers. For example, in response to a UE context release the server may destroy one or more containers. Also for example, in response to an initial UE attach, the server may create one or more containers. In still further examples, if a serving or packet gateway relocation results from a handover, the server may move a container from one edge cloud location to another edge cloud location. These and other embodiments are described further below.

(68) FIG. 2H is a block diagram illustrating an example, non-limiting embodiment of an application container at a first edge cloud location in accordance with various aspects described herein. FIG. 2H shows UE **210A**, application **212H**, radio access network node **112H**, network core **125**, edge cloud location **250H** with application container **252H**, network API service **158**, and

resource manager **160H**. UE **210A** is described above with reference to previous figures.

Application **212H** may be any application running on UE **210A**. For example, application **212H** may be a video application on a mobile device, an entertainment application within an automobile, an application running on a laptop device connected to a Wi-Fi access point, or the like.

(69) Radio access network node **122H** may be any radio access network node capable of communicating with UE **210A**. For example, in some embodiments, radio access network node **122H** is one of the radio access network nodes described with reference to FIG. **1**. In some embodiments, radio access network node **122H** may be a 4G radio access network node, and in other embodiments, radio access network node **122H** may be a 5G radio access network node.

(70) Network core **125** includes any of the network elements described with reference to FIG. **1**. In general, network core **125** may be any network capable of providing services to UE **210A** through radio access network node **122H**.

(71) Edge cloud location **250H** is one of many possible edge cloud locations. For example, an edge cloud platform provider may provide for many different possible edge cloud locations at which application containers may be instantiated. For example, various edge cloud locations may be physically located near serving gateways, packet gateways, or other user plane functions through which data is routed. Utilizing edge cloud locations in physical proximity to these gateways or user plane functions may provide for decreased latency and/or increased functionality in general.

(72) Application container **252H** is an application container running at edge cloud location **250H**. For example, edge cloud location **250H** may include any combination of physical servers and virtual servers, and application container **252H** may be container running on any one or multiple of these servers. In some embodiments, application container **252H** is specific to a user. For example, application container **252H** may house one or more applications specific to a user of UE **210A**. Also for example, in some embodiments, application container **252H** may be specific to an application or an application type. For example, application container **252H** may include applications that communicate with application **212H** and/or a multiple instantiations of application **212H**.

(73) Network API service **158** receives network event data from various entities and provides performance hints as described above. For example, in some embodiments, network API service **158** may receive network event data from an access network that includes radio access network node **122H**. In some embodiments, network API service **158** receives network event data directly from radio access network node **122H**. Also for example, network API service **158** may receive network event data from network elements within network core **125**. In response to network event data received from various elements within the network, network API service **158** may provide performance hints to resource manager **160H**.

(74) Resource manager **160H** is an entity that manages resources within various edge cloud locations. For example, in some embodiments, resource manager **160H** is provided by an edge cloud service provider and manages many or all functions having to do with network services provided at network edge locations. For example, an edge cloud platform resource manager may boot application containers in one or many different edge cloud locations corresponding to services provided by the edge cloud platform service provider. Also for example, resource manager **160H** may be an edge application master node which manages all processes associated with a particular application. For example, application **212H** may be a video application, and resource manager **160H** may manage all application containers instantiated at edge cloud locations that provide services to application **212H**. Resource manager **160H** may be any type of entity or process capable of utilizing performance hints provided by network API service **158** in the management of application containers at edge cloud locations.

(75) An edge cloud platform may include of any number of edge cloud locations implemented in small data centers deployed at hundreds of locations within a network. For example, an edge cloud platform may include many geographically distributed edge cloud locations such as edge cloud

location **250H**. Resource allocation may be challenging at an edge cloud location in part because of their small size and the utility of supporting a variety of applications. In some embodiments, a large number of applications may be supported using containers or lightweight virtual machines that consume far fewer resources than having a dedicated server or a traditional virtual machine for each application; however, it may not be possible to simultaneously run all application containers for all users due to limited resources. In addition, since a user may use an application only a fraction of the time, constantly running application containers for all users and/or applications may be wasteful of compute resources. Various embodiments described herein may provision application containers “just-in-time” upon the arrival of a new request from a user. This saves compute resources and is relatively fast since containers can be started or stopped quicker than virtual machines; however just-in-time provisioning may still have a noticeable impact on user-perceived response time while the container boots up for a few seconds. Further embodiments described herein may reduce the impact on user-perceived response times in part by predicting via network event data that an application may be used before the arrival of a user request.

(76) FIG. 2I is a block diagram illustrating an example, non-limiting embodiment of an application container at a first edge cloud location in accordance with various aspects described herein. FIG. 2I shows UE **210A** communicating with radio access network node **122H**. FIG. 2I also shows radio access network node **122H** communicating with application container **252H** at edge cloud location **250H** through a network element **110I** which is part of network core **125**. Network element **110I** may be any network element capable of providing a data path between radio access network node **122H** and edge cloud location **250H**. For example, network element **110I** may be a user plane function, a serving gateway, a packet gateway, or the like. Communications between radio access network node **122H** and edge cloud location **250H** may in general take any path through network core **125** without limitation. FIG. 2I also shows radio access network node **122I**, network element **112I**, and edge cloud location **250I**. In embodiments represented by FIG. 2I, UE **210A** is located physically such that it communicates with radio access network node **122H** rather than network access node **122I**. In some embodiments, network API service **158** (not shown) receives network event data from one or both of radio access network node **122H** and network core **125** and provides performance hints to a resource manager that boots application container **252H** at edge cloud location **250H**.

(77) FIG. 2J is a block diagram illustrating an example, non-limiting embodiment of an application container moving from a first edge cloud location to a second edge cloud location in accordance with various aspects described herein. FIG. 2J shows the same elements as FIG. 2I, with the exception that UE **210A** has been handed over from radio access network node **122H** to radio access network node **122I**. In some embodiments, this corresponds to UE **210A** physically moving and the radio access network recognizing a location change and handing over communications from radio access network node **122H** to radio access network node **122I**. In some embodiments, in response to network event data resulting from the handover shown in FIG. 2J, network API service **158** (not shown) provides performance hints to a resource manager, and the resource manager takes down the application container that was running at edge cloud location **250H** and boots an application container **252J** at edge cloud location **250I**. As shown in FIG. 2J, the communications path is now from UE **210A** through radio access network node **122I** and network element **112I** to application container **252J** running at edge cloud location **250I**.

(78) Performance hints provided by network API service **158** may be used for any purpose, including booting up application containers as well as shutting down or deleting application containers. For example, when a user disconnects from the network or its network traffic become idle, the network API service may hint the edge cloud to delete one or more application containers. An example of such an event from the radio access network is an RRC_IDLE event, which shows that user has been idle for last X seconds. In some embodiments, a lack of network traffic may also be reported by a packet gateway (5G UPF or 4G SAEGW) that may process all traffic. Further,

mobility management (4G MME, or 5G AMF) may report if a user has explicitly deregistered from the network or its connection is terminated by the network. Using these hints to deallocate resources may free up resources at an edge cloud to potentially serve other active users and their applications.

(79) FIG. 2K is a block diagram illustrating an example, non-limiting embodiment of an edge cloud platform locating an application server in an edge cloud location that communicates through a local breakout function in accordance with various aspects described herein. Similar to FIG. 2J, FIG. 2K shows a hand over of UE **210A** between radio access network nodes. In the example of FIG. 2K, the radio access network has handed off UE **210A** from radio access network node **122H** to radio access network node **122K**. Radio access network node **122K** communicates with edge cloud location **250K** through a local breakout **112K**. In some embodiments, a local breakout **120K** provides a more efficient path to an edge cloud location rather than going through an entire network core such as network core **125**. In some embodiments, a local breakout such as local breakout **112K** may be a lightweight network function within network core **125** or may bypass network core **125** altogether to provide a more efficient path to an edge cloud location such as edge cloud location **250K**. In response to network event data received as part of the handover, a network API service may provide performance hints to a resource manager that may tear down an application container at edge cloud location **250H** and instantiate application container **252K** at edge cloud location **250K**.

(80) Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular, a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of the systems and methods presented in previous figures. For example, virtualized communication network **300** can facilitate in whole or in part a network API service that combines data from access and core networks and provides guidance and control services to API users.

(81) In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

(82) In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general purpose processors or general purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

(83) As an example, a traditional network element **150** (shown in FIG. 1), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it's elastic: so the

resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle-boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

(84) In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized, and might require special DSP code and analog front-ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

(85) The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network Elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements don't typically need to forward large amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and overall which creates an elastic function with higher availability than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

(86) The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud, or might simply orchestrate workloads supported entirely in NFV infrastructure from these third party locations.

(87) Turning now to FIG. **4**, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **4** and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330**, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part a network API service that combines data from access and core networks and provides guidance and control services to API users.

(88) Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers,

mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

(89) As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

(90) The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

(91) Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

(92) Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

(93) Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

(94) Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

(95) With reference again to FIG. 4, the example environment can comprise a computer **402**, the computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **404**.

(96) The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any

of a variety of commercially available bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

(97) The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

(98) The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

(99) A number of program modules can be stored in the drives and RAM **412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

(100) A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

(101) A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

(102) The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements

described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

(103) When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

(104) When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

(105) The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

(106) Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

(107) Turning now to FIG. 5, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, platform **510** can facilitate in whole or in part a network API service that combines data from access and core networks and provides guidance and control services to API users. In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks, and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway

node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

(108) In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

(109) In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

(110) For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in

FIG. 1 that enhance wireless service coverage by providing more network coverage.

(111) It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processors can execute code instructions stored in memory **530**, for example. It should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

(112) In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

(113) In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

(114) Turning now to FIG. 6, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part a network API service that combines data from access and core networks and provides guidance and control services to API users.

(115) The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, WiFi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1×, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VOIP, etc.), and combinations thereof.

(116) The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608**

can be presented by way of the display **610** with navigation features.

(117) The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

(118) The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

(119) The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

(120) The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

(121) The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, WiFi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

(122) Other components not shown in FIG. **6** can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

(123) The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn't otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice

versa, etc.

(124) In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, nonvolatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

(125) Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

(126) In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

(127) Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $x=(x_1, x_2, x_3, x_4, \dots, x_n)$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches

comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

(128) As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

(129) As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

(130) Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

(131) In addition, the words “example” and “exemplary” are used herein to mean serving as an

instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

(132) Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

(133) Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

(134) As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

(135) As used herein, terms such as “data storage,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

(136) What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

(137) In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

(138) As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

(139) Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

Claims

1. A method comprising: receiving, by a processing system including a processor, network event data at a network application programming interface (API) from a communication network; in response to the network event data, predicting, by the processing system, operations that may enhance performance of edge cloud entities in communication with the communication network; and providing, by the processing system, via the network API, a response indicative of performance hints, wherein the providing the performance hints comprises providing actions or commands that enhance performance of edge cloud entities.
2. The method of claim 1, wherein the providing the performance hints further comprises: providing, by the processing system, identification information for edge cloud locations at which a first application container is available for commands.
3. The method of claim 2, wherein the providing the performance hints further comprises: providing, by the processing system, commands to boot a second application container or a first virtual machine at first edge cloud location; and providing, by the processing system, commands to take down a third application container or a second virtual machine at a second edge cloud

location.

4. The method of claim 1, further comprising: registering, by the processing system, the network API with the communications network to receive the network event data at the network API.

5. The method of claim 4, wherein the registering the network API comprises: registering, by the processing system, the network API to receive the network event data from a first plurality of network elements of an access network of the communication network; and registering, by the processing system, the network API to receive the network event data from a second plurality of network elements within a core network of the communication network.

6. The method of claim 5, further comprising: predicting, by the processing system, ingestion of data streams of the network event data from the first plurality of network elements, the second plurality of network elements, or both.

7. The method of claim 1, further comprising: analyzing, by the processing system, the network event data; predicting, by the processing system, the performance hints that enhance performance of edge cloud entities; and providing, by the processing system, the performance hints to server applications.

8. The method of claim 7, further comprising: providing, by the processing system, a list of network APIs available to the server applications at a network API gateway.

9. The method of claim 1, further comprising: receiving, at the network API, a request to authenticate, on behalf of a user, on behalf of a user equipment (UE) device, or on behalf of an application running on the UE device, to determine different network API service privileges associated with the user, the UE device, or the application.

10. The method of claim 9, further comprising: determining, by the processing system, that the UE device has network API service privileges by verifying that the UE device is registered to a user of a particular service provider.

11. A device, comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: receiving first network event data at a network application programming interface (API) from a core network of a communication network; receiving second network event data at the network API from an access network of the communication network; and providing, via the network API, information including performance hints based on the first network event data and the second network event data, wherein the providing the information including the performance hints includes providing information about actions or commands targeted to enhance performance of edge cloud entities.

12. The device of claim 11, wherein the operations further comprise: registering the network API with the communication network to receive the first network event data and the second network event data.

13. The device of claim 11, wherein the operations further comprise: predicting, at the network API, ingestion of data streams of the first network event data from network elements of the core network, ingestion of data streams of the second network event data from network elements of the access network, or both; and providing, at the network API, edge cloud related hints to allow an edge cloud resource manager associated with the communication network to boot one or more application containers in an efficient manner.

14. The device of claim 11, wherein the operations further comprise: analyzing the first network event data and the second network event data; predicting the performance hints to enhance performance of the edge cloud entities; and providing the performance hints to server applications.

15. The device of claim 14, wherein the operations further comprise: providing a list of network APIs available to the server applications at a network API gateway of the communication network.

16. The device of claim 14, wherein the predicting the performance hints comprises: predicting future behavior of user equipment in the communication network based on the first network event data, the second network event data and stored past network event data.

17. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: registering a network application programming interface (API) with a communications network; receiving first network event data at the network API from a core network of the communication network; receiving second network event data at the network API from an access network of the communication network; receiving, from a user equipment device of the communication network, an authentication request; authenticating the user equipment device; receiving, from a server application, a request for performance hints; and in response to the request for performance hints, providing, by the processing system, via the network API, a response including information about the performance hints, the performance hints developed based on the first network event data and the second network event data, wherein the information about the performance hints includes information about actions or commands that enhance performance of edge cloud entities associated with the communication network.

18. The non-transitory machine-readable medium of claim 17, wherein the providing the response comprises: subscribing, by the network API, to one or more services published by one or more elements within the access network, within the core network, or both; and receiving, at the network API, the first network event data and the second network event data published via the one or more services.

19. The non-transitory machine-readable medium of claim 17, wherein the operations further comprise: receiving, at the network API, raw data streams from one or more network elements within the access network, the core network, or both; and based on the raw data streams, producing the performance hints substantially in real time.

20. The non-transitory machine-readable medium of claim 17, wherein the operations further comprise: receiving at the network API, from a geographical region of the communication network, a request for services; and redirecting the request for services to another network API that is physically located in the geographical region where the request for services originates.
